

GAIA: Glass-Aware I/O Middleware

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Abstract—As cloud-scale services and data-centric applications continue to generate massive volumes of data, the need for ultra-durable, energy-efficient, and cost-effective archival storage becomes increasingly urgent. Quartz glass has recently emerged as a promising archival medium, offering multi-century durability, radiation and thermal resistance, and support for three-dimensional data encoding using femtosecond laser writing. However, the hybrid mechanical-optical architecture of glass storage—requiring mechanical movement along the X and Y axes and optical focal tuning along the Z axis—introduces unique performance bottlenecks during data access, which conventional I/O scheduling strategies are not equipped to handle.

In this work, we present GAIA, a Glass-Aware I/O middleware designed to optimize data access in quartz glass storage systems. GAIA features three coordinated strategies: (1) Zigzag Data Placement, which aligns data with the mechanical stage’s natural motion to minimize direction-switching latency; (2) Z-Axis First Placement, which prioritizes low-latency optical traversal along the depth dimension; and (3) Shortest Moving Time First (SMTF) scheduling, which selects I/O operations based on predicted movement time rather than geometric distance. Through trace-driven simulations using enterprise-scale workloads and various glass sizes, GAIA reduces data read latency by up to 82% compared to traditional baseline schedulers. These results demonstrate the critical importance of middleware-level co-design in unlocking the performance potential of next-generation glass-based storage systems.

I. INTRODUCTION

Driven by the explosive growth of cloud-scale services and data-centric applications [1], the demand for ultra-durable, cost-effective archival storage continues to surge [2], [3], [4], [5]. Applications ranging from GitHub’s Arctic Code Vault [6], and Google Cloud’s audit and compliance logging [7], to long-term scientific repositories like CERN’s Open Data Portal [8], NASA’s Earth science archive [9], and Amazon Aurora’s historical database snapshots, now require storage media capable of preserving data reliably for decades or even centuries. Traditional storage media—such as magnetic tapes and hard disk drives (HDDs)—are increasingly challenged by physical wear, bit rot, and environmental degradation, limiting their long-term sustainability [10], [11], [12], [13]. To address this, quartz glass has emerged as a compelling storage medium, offering exceptional durability, radiation and heat resistance, and the

ability to store data in three dimensions with femtosecond laser precision [14], [15], [16], [17]. These properties make it especially promising for cold storage in cloud and enterprise systems. However, despite its durability and density, quartz glass storage introduces unique performance bottlenecks during data access due to its mechanical-optical architecture. In particular, the need for high-precision stage movement in the X-Y plane and optical focal tuning along the Z-axis results in nontrivial access latency that current I/O schedulers cannot handle properly.

The growing industry investment in long-term archival storage further emphasizes the critical need for high-density and stable solutions. Notable examples include Microsoft’s Project Silica [18], [19], which pioneers glass-based storage systems for data centers, and broader optical research into multi-layer and multi-dimensional encoding [16]. These works illustrate the relevance and potential of glass storage for cloud-scale workloads. Moreover, regulatory and scientific domains increasingly demand immutable and audit-friendly data preservation formats, such as Write-Once-Read-Many (WORM) media. Despite the emergence of promising hardware techniques for writing and retrieving data in quartz glass, current systems still treat the medium as a passive archival layer, lacking middleware support to manage access efficiency at scale. This mismatch motivates the need for a system-level rethinking of data placement and I/O scheduling tailored to the physical properties of quartz-based media.

Several foundational works have demonstrated the feasibility of writing data to quartz glass using femtosecond laser pulses and reading it back through polarization-sensitive microscopy [20], [16], [18]. These techniques enable encoding multiple bits per voxel and support three-dimensional, layer-by-layer data placement, paving the way for exabyte-scale archival solutions. However, most existing efforts focus on hardware-level innovations, and the system-level impact of mechanical stage dynamics and optical adjustments remains underexplored. For example, the delays introduced by mechanical acceleration, directional switching, and Z-axis scanning are not addressed in conventional disk or flash-based I/O

schedulers [21], [22], [23], [24]. As a result, glass storage systems lack practical data management policies that bridge the gap between their unique physical access model and modern data service requirements.

In this work, we present GAIA, a novel Glass-Aware I/O middleAre, designed to optimize read performance in 3D glass storage systems. We begin by revisiting conventional storage scheduling techniques and adapting them to the hybrid mechanical-optical characteristics of glass-based access. Our middleware introduces three core components. First, we propose Zigzag Data Placement, a planar data layout that minimizes X-Y stage resets by scanning blocks in a directional-preserving zigzag pattern. Second, we introduce Z-Axis First Data Placement, which prioritizes vertical layout to exploit low-latency electronic focal tuning and reduce horizontal movement overhead. Third, we present Shortest Moving Time First (SMTF) scheduling, an acceleration-aware algorithm that minimizes actual seek time by accounting for axis-specific speed, direction-switching latency, and mechanical inertia. Through a combination of simulated enterprise-scale workloads and realistic hardware parameters, our evaluation shows that this middleware significantly improves read throughput, achieving up to 82% latency reduction compared to traditional distance-based schedulers. These results demonstrate the critical role of system-level co-design in unlocking the performance potential of quartz glass storage for future archival applications.

II. BACKGROUND AND MOTIVATION

A. Background

1) *Architecture of Glass Storage:* Glass-based data storage leverages femtosecond laser technology to write information directly into quartz glass, a highly durable and optically transparent medium. During the writing process, an ultra-short, high-intensity laser pulse is precisely focused into the interior of the glass, where it induces localized physical modifications to the material's structure—effectively encoding data within the glass volume itself. Modern glass storage systems adopt an architecture that separates the data writing and data reading processes into two independent components, each specialized for its role in the storage pipeline [25], [19].

The writing component utilizes a polarized femtosecond laser beam, directed into the quartz glass through a high-precision optical system. When focused, the laser produces micro-explosions or multiphoton absorption events that create voxel-shaped structural changes at targeted depths within the glass matrix [26]. These voxels serve as the basic units of data, encoding bits through properties such as birefringence, retardance, and polarization angle [16], [27]. The precision of this technique enables multi-dimensional encoding and high data density, with each voxel storing multiple bits.

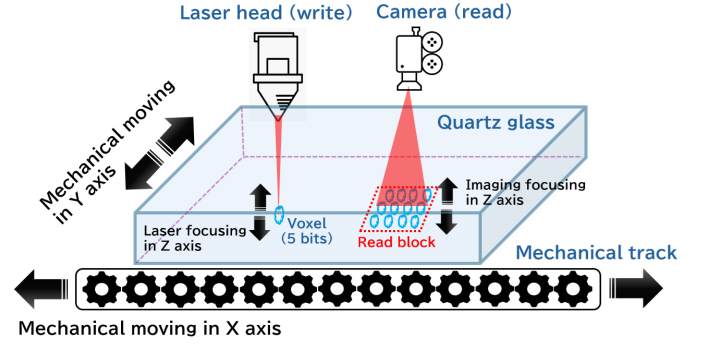


Fig. 1. Architecture of Glass Storage: Separation of Write and Read Devices

The reading component is designed to retrieve the encoded information by analyzing the optical characteristics of each voxel. Using techniques such as polarization microscopy or interferometry, the system detects variations in birefringence and orientation, which are mapped back to the original bit values [18]. A more detailed explanation of the bit-level decoding methodology is provided in Section II-A2.

In current femtosecond laser writing systems, quartz glass is mounted on a motorized sample stage that moves along X and Y axes via mechanical tracks (as illustrated in Figure 1). This mechanical movement introduces a significant performance bottleneck during data writing, as repositioning delays dominate the overall latency. In contrast, Z-axis focal adjustments are performed electronically, offering much lower latency. Notably, both writing and reading devices rely on this mechanical track system to reposition the sample beneath the laser or sensor.

While previous studies have demonstrated rewrite capabilities in quartz glass [20], [16], the current primary application remains in archival storage for infrequently accessed data, such as that used in cloud-scale archival systems [19]. Due to the high time cost of rewriting, glass storage today is best suited for write-once-read-many (WORM) scenarios [18].

2) *How to Read From Quartz Glass Storage Devices:* Data decoding in glass storage is achieved by imaging the internal three-dimensional voxel structures using a polarization optical microscope. This process captures subtle optical modifications introduced during the femtosecond laser writing phase. Specifically, femtosecond pulses induce localized structural changes within the quartz glass, such as form birefringence and retardance, which modulate the polarization of light passing through the affected regions [18]. These optical properties, though invisible under conventional lighting conditions, become highly distinguishable when viewed through a polarization-sensitive imaging system, thereby enabling non-destructive readout of embedded data. The precise modulation

of refractive indices within each voxel is what makes data storage in quartz glass both high-density and resilient to environmental degradation.

Each read operation captures a two-dimensional planar region, commonly referred to as a read block, which contains an array of individually written voxels. Each voxel, acting as a discrete unit of information, typically encodes up to 5 bits through multi-level encoding mechanisms, such as orientation and strength of birefringence [16]. The number of voxels per read block is determined by the spatial resolution and magnification of the polarization microscope, and by the spacing precision achieved during the laser writing process. These read blocks are distributed across multiple Z-levels within the glass substrate, resulting in a fully volumetric data layout. In contrast to traditional two-dimensional storage media, glass-based systems exploit the depth dimension to increase capacity and reduce surface congestion, enabling multi-layer encoding with gigabit- to terabit-scale densities per platter.

As discussed in Section II-A1, mechanical movement of the glass substrate across the X and Y axes is managed by independent motorized linear track systems. These systems translate the sample stage beneath the fixed microscope objective in a sequential, axis-aligned fashion—typically following a Manhattan-distance traversal model, where motion occurs first along the X direction and then along Y. However, because these mechanical components are subject to physical acceleration and deceleration limits, frequent start-stop motion and direction switches introduce substantial overhead. The time it takes for the mechanical stage to accelerate and decelerate when shifting between distant read blocks in the X-Y plane is a major contributor to overall data access latency. Consequently, read scheduling strategies must be acutely aware of stage dynamics, spatial layout, and movement history to minimize unnecessary repositioning.

In contrast to the relatively slow mechanical traversal along X and Y, vertical movement along the Z-axis is achieved electronically by adjusting the focal plane of the polarization optical microscope. This adjustment is controlled via piezoelectric actuators or precision lens-shifting mechanisms, allowing for rapid, low-latency refocusing between adjacent Z layers without any physical movement of the stage or optical head. As a result, Z-axis traversal can be executed orders of magnitude faster than planar movement, enabling high-throughput, layer-by-layer decoding of data. This performance asymmetry strongly motivates the design of data layouts and access patterns that prioritize Z-axis continuity. Read scheduling strategies that favor depth-wise scanning—by minimizing horizontal repositioning and maximizing consecutive Z-layer access—can thus significantly improve overall decoding efficiency and reduce the time required for bulk read operations in quartz glass storage systems.

B. Motivation

To the best of our knowledge, no prior work has proposed an I/O scheduling strategy specifically optimized for glass-based data storage. Traditional scheduling algorithms widely used in conventional storage systems, such as Shortest Seek Distance First (SSDF) and shortest path algorithms [28], [29], have not been adapted to exploit the unique architectural and physical characteristics of glass storage, as described in Section II-A.

When applied in a three-dimensional Manhattan space, neither SSDF nor shortest path algorithms effectively address the acceleration and deceleration dynamics inherent to the X and Y mechanical axes of glass storage systems. These legacy algorithms are primarily designed for rotational media like HDDs, where seek latency is a function of angular distance rather than the physical inertia of linear mechanical stages. As a result, they overlook the directional switching penalties and fail to optimize for smooth, continuous motion—a critical factor in glass-based systems where repositioning delays dominate access time.

Additionally, existing approaches do not leverage the asymmetry in axis efficiency: in glass storage, Z-axis access via optical focal adjustment is significantly faster than X or Y axis movement. This underutilized Z-axis advantage represents a missed opportunity for performance gains, particularly in 3D data layouts where vertical traversal can be prioritized.

While acceleration-aware scheduling may yield longer Manhattan distances, it compensates by minimizing frequent starts and stops, thus reducing total access time. By avoiding unnecessary directional changes and enabling longer, uninterrupted movement across the X and Y axes, such strategies better align with the physical realities of the system. This limitation of SSDF becomes particularly apparent in devices where mechanical inertia and stage acceleration latency are significant performance factors.

Furthermore, cloud-scale storage environments often impose restrictions on I/O request queue lengths, limiting the scope for global reordering and optimization [30], [31]. As such, there is a clear need for a glass-aware scheduling middleware layer that is not only system-aware, but also explicitly tailored to the physical traits of the storage medium. This middleware must operate under practical constraints while intelligently scheduling read operations to maximize efficiency across all three spatial dimensions.

III. GAIA:GLASS-AWARE I/O MIDDLEWARE

A. Overview

In this work, we propose GAIA: Glass-Aware I/O Middleware, a comprehensive system framework designed to improve data access efficiency in quartz glass storage systems. Unlike conventional storage devices, glass storage operates in a three-dimensional physical domain, where data is written and read

by controlling both mechanical movement along the X-Y plane and optical focal adjustments along the Z-axis. This hybrid mechanism introduces a significant performance asymmetry: while X-Y movements are limited by mechanical acceleration and deceleration, Z-axis access is executed electronically and can be completed at much higher speeds. Existing I/O scheduling techniques, such as those developed for hard drives (FCFS and SSTF) [32], do not consider this asymmetry and therefore fail to minimize the true cost of access in glass storage. To address this, GAIA introduces a middleware layer that redefines data placement and scheduling policies with awareness of the underlying physical dynamics.

GAIA comprises three key strategies:

- **Zigzag Data Placement:** Improves planar read efficiency by reducing idle time and minimizing backtracking during data access. This strategy arranges read blocks across the X-Y plane in a zigzag pattern, allowing the mechanical stage to scan them sequentially in alternating directions. By aligning data placement with the natural movement path of the stage, it avoids frequent long-distance resets and adheres to the physical constraints of quartz glass storage, thereby enhancing overall read throughput.
- **Z-Axis First Data Placement:** Z-Axis First Data Placement: Organizes read blocks in a Z-major order, prioritizing vertical (Z-axis) layering over traditional X-Y planar layouts. This strategy takes advantage of the low-latency electronic focal adjustments used for Z-axis access, enabling fast, layer-by-layer reading without the need for frequent mechanical repositioning. By minimizing X-Y stage movements, it significantly reduces overall access latency and improves efficiency in reading three-dimensional data structures within quartz glass..
- **Shortest Moving Time First (SMTF):** A novel scheduling algorithm that extends beyond traditional distance-based heuristics. SMTF considers mechanical acceleration and deceleration behavior, prioritizing read requests that result in the shortest actual movement time. It accounts for axis-specific speed and direction-switching overheads to make scheduling decisions that minimize total read latency.

Our middleware integrates these strategies into a unified scheduling layer that operates under realistic hardware constraints, such as limited queue capacity and non-preemptive mechanical stages. It dynamically adapts to runtime access patterns and current stage positions, selecting the most suitable strategy to maintain high throughput and low latency. This design enables practical and scalable deployment of quartz glass storage systems in data-intensive environments.

B. Glass-Oriented Data Placement

1) **Zigzag Data Placement Strategy:** In large-scale cloud storage systems, each read request typically consists of several

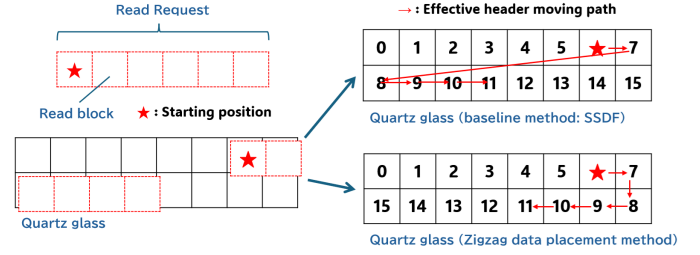


Fig. 2. Zigzag data placement improves stage traversal efficiency by minimizing directional changes compared to SSDF.

consecutive read blocks. These blocks are located at continuous physical addresses. They are always accessed in ascending order, from the lowest to the highest address, as shown in the left half of Figure 2.

When a straightforward SSDF strategy is applied, the continuity of address-based access results in an inefficiency: upon reaching the read block located at the far right (the tail) of the glass, the read head must return to the far left (the head) to access the next block. This leads to a long and inefficient traversal, as illustrated in the upper-right part of Figure 2 (baseline method: SSDF). The mismatch between data layout and physical access direction causes excessive seek distances and frequent directional changes.

To mitigate this inefficiency, we introduce a zigzag data placement mechanism. In this design, data blocks are pre-arranged in a spatially continuous zigzag pattern on the surface of the quartz glass. As shown in the lower-right part of Figure 2, the logical order of blocks (from 0 to 15) is mapped to a physical layout that enables zigzag traversal. Instead of resetting to the beginning of the next row—as in conventional row-major ordering—the read head proceeds in the opposite direction at the end of each row.

By aligning the physical layout of data with the natural scan path of the mechanical stage, the zigzag method reduces the number of directional changes and the total seek time. It minimizes acceleration overhead, particularly in the X-Y plane.

2) **Z-axis First Data Placement Strategy:** In quartz glass storage systems, the Z-axis offers a significant performance advantage due to its distinct access mechanism. Unlike the X and Y axes—which rely on mechanical translation of the stage—the Z-axis is adjusted electronically via focal tuning of the polarization optical microscope. This fundamental difference leads to a stark contrast in access speed: while the mechanical stage reaches a maximum speed of approximately 800 mm/s in the X-Y plane [33], Z-axis focal adjustments can effectively switch between layers at speeds equivalent to 15 m/s, nearly 20× faster [34].

This disparity in access latency directly motivates the Z-

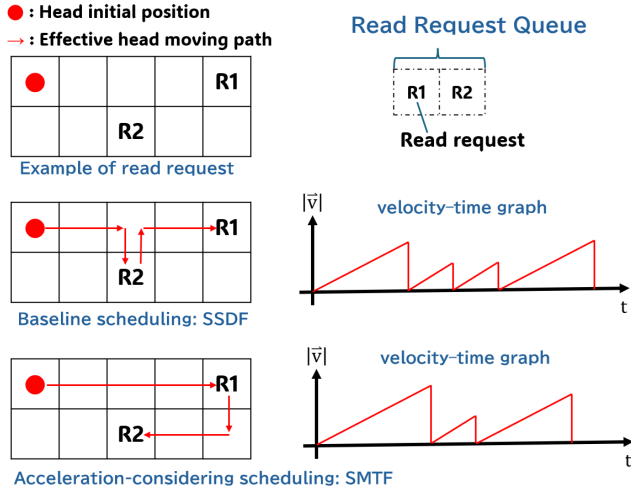


Fig. 3. Comparison between SSDF and SMTF scheduling strategies. SMTF outperforms SSDF by accounting for acceleration and deceleration overheads in mechanical stage movement, resulting in reduced total seek time.

Axis First Data Placement strategy. By arranging read blocks in a Z-major order, the system can perform rapid, layer-by-layer scanning using optical focal adjustments with minimal or no mechanical delay. Data is first filled along the depth (Z-axis) before extending across the lateral X and Y dimensions. This layout enables the scheduler to exploit the low-latency Z traversal for sequential reading, avoiding frequent and costly repositioning of the X-Y stage.

Additionally, Z-axis-first placement aligns well with read-intensive workloads that favor 3D spatial locality, allowing multiple consecutive read requests to be satisfied within the same X-Y coordinate by simply sweeping through successive Z layers. This minimizes back-and-forth mechanical motion and maximizes temporal efficiency, making it particularly effective under hardware constraints such as non-preemptive stage movement and limited I/O request queue sizes.

In summary, by prioritizing data layout along the Z dimension, this strategy leverages the asymmetry in axis latency to accelerate access, reduce mechanical wear, and improve overall throughput in quartz glass storage systems.

C. Shortest Moving Time First (SMTF) I/O Scheduler

To optimize read efficiency in quartz glass storage systems, we progressively refine the I/O scheduling strategy to better reflect the underlying physical characteristics of the hardware. We begin with the First-Come, First-Served (FCFS) approach as a brute-force baseline. FCFS processes requests strictly in the order of arrival without any awareness of spatial locality. While simple, this strategy often results in inefficient mechanical movement, particularly when access patterns are scattered or non-contiguous.

To improve spatial awareness, we next introduce Shortest Seek Distance First (SSDF), which selects the next read request based on the shortest Manhattan distance in the X-Y plane. This method aims to reduce overall mechanical translation by minimizing the linear distance between the current stage position and the target block. However, while SSDF can reduce aggregate movement distance, it overlooks an important physical limitation of the system: the mechanical stage incurs additional latency due to acceleration, deceleration, and direction changes. Frequent switches between X and Y axes or abrupt reversals in movement direction lead to increased seek time, even when total distance is low. As a result, SSDF does not necessarily minimize total access latency, especially in workloads that exhibit localized or sequential access patterns.

Building on this foundation, we propose Shortest Moving Time First (SMTF), a more advanced and physically-aware scheduling strategy. Unlike SSDF, SMTF explicitly models the movement behavior of the mechanical stage, including axis-specific acceleration profiles, direction-switching penalties, and differences in movement efficiency between X and Y axes. Rather than selecting the request closest in distance, SMTF prioritizes the request that can be completed in the shortest actual movement time, taking into account both spatial layout and motion dynamics. This allows the stage to maintain longer uninterrupted motion segments and reduces latency associated with stop-and-go movements.

SMTF is designed to operate efficiently under practical hardware constraints, such as non-preemptive stage control and bounded I/O queue sizes. Unlike traditional distance-based strategies, SMTF optimizes for actual movement time rather than geometric proximity alone. By accounting for acceleration and deceleration dynamics, it enables smoother and more stable access patterns, leading to improved read throughput and reduced latency in real-world quartz glass storage systems.

To illustrate the advantage of SMTF over SSDF, Figure 3 presents a toy example comparing the two strategies. Given the current position of the read head and two upcoming read requests, R1 and R2, we plot the velocity-time profiles for both SSDF and SMTF. While SMTF may result in a slightly longer Manhattan distance, it allows the mechanical stage to maintain a higher sustained velocity by reducing abrupt directional changes. Moreover, with SMTF the stage reaches a higher peak velocity by better leveraging the acceleration phase. Thus, SMTF completes the requests earlier than SSDF. This demonstrates the advantage of acceleration-aware scheduling, where optimizing for movement time, not just distance, leads to more efficient execution.

D. Put Them All Together

Figure 4 summarizes the evolution of our scheduling strategies, starting from conventional approaches and progressively

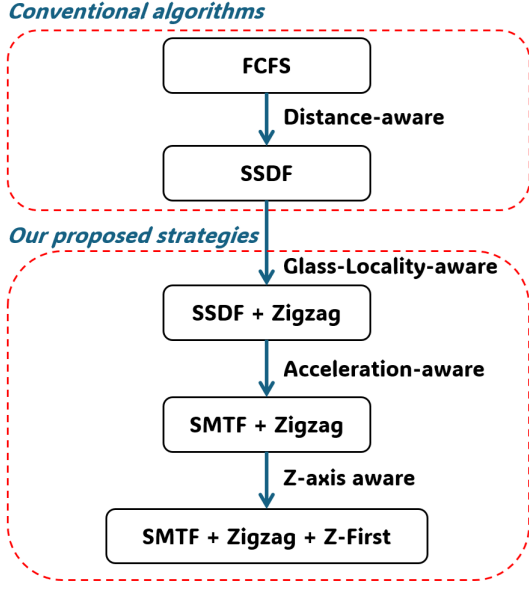


Fig. 4. Evolution of I/O scheduling strategies, from conventional algorithms to our proposed glass-aware, acceleration-aware, and Z-axis-aware methods.

incorporating the physical and architectural characteristics of quartz glass storage. At the baseline, the First-Come, First-Served (FCFS) algorithm processes read requests strictly in arrival order, without considering spatial locality. Shortest Seek Distance First (SSDF) improves upon FCFS by minimizing the total movement in the X-Y plane based on Manhattan distance, making it distance-aware. However, SSDF still falls short in real-world quartz storage systems due to its disregard for acceleration penalties and mechanical movement behavior.

To address this, we first enhance SSDF with our Zigzag data placement strategy, which improves X-Y stage efficiency by reducing abrupt direction changes, making it glass-locality-aware. We then propose the Shortest Moving Time First (SMTF) scheduler, which, when combined with Zigzag placement, becomes acceleration-aware by optimizing for actual movement time rather than geometric proximity alone. This allows the stage to maintain smoother, longer motion segments, enabling SMTF to complete requests earlier than SSDF despite sometimes traveling a longer distance.

Finally, we integrate Z-axis prioritization into our scheduling stack. By combining SMTF with Zigzag and Z-axis First placement, the strategy becomes fully aware of the quartz glass storage’s 3D access behavior. This end-to-end design leverages fast Z-axis optical adjustments, acceleration-aware motion planning, and spatially optimized data placement to significantly reduce overall seek time and improve read throughput in quartz-based archival systems.

IV. PERFORMANCE EVALUATION

A. Performance Metrics and Evaluation Setup

In our evaluation, total latency is used as the primary performance metric, as it accurately reflects the true cost of accessing read blocks across a three-dimensional glass storage medium. This metric captures both mechanical movement delays and optical focal adjustment times, offering a comprehensive view of system performance under various scheduling strategies.

To simulate realistic workloads, we adopt an I/O trace dataset from Fujitsu [35], [36], which records enterprise-scale cloud storage activity. This trace serves as the source of incoming read requests in our simulation, enabling us to evaluate performance under representative, real-world access patterns.

We develop a trace-based simulation framework that emulates the behavior of a quartz glass storage system at the middleware level. This framework includes a read request queue, a scheduling module, and detailed logic to model stage movement and optical focal adjustment. The simulator tracks physical constraints such as acceleration, directional changes, and axis-specific latency to ensure fidelity to actual hardware behavior.

Due to the limited capacity of the request queue, only a portion of the dataset is available to the scheduler at any given moment. As a result, the scheduling algorithm operates under partial knowledge, making it infeasible to compute globally optimal schedules. Nonetheless, our middleware enables adaptive, near-optimal scheduling decisions by exploiting local access patterns and physical constraints—offering a practical and scalable solution for real-world deployment.

In our simulation of glass sizes and their corresponding data capacities, we refer to the voxel spacing specifications presented in the literature [18]. The size of the glass medium significantly affects the performance of I/O scheduling strategies, especially in systems that involve mechanical movement and 3D data access. To evaluate the effectiveness of our proposed strategies under realistic constraints, we simulate three representative glass sizes: small, medium, and large. The small glass configuration consists of $32 \times 16 \times 10$ read blocks across layers, supporting approximately 20.5 MB of data. The medium glass is composed of $160 \times 80 \times 100$ read blocks, with a total data capacity of 5.12 GB. The large glass contains $640 \times 320 \times 1000$ read blocks, accommodating up to 820 GB of data. These configurations reflect practical design assumptions and allow us to assess how scheduling strategies scale with data volume and accessing performance.

B. Read-Write Head Simulation

To estimate the seek latency of the mechanical stage in glass storage, we adopt a two-phase physical movement model inspired by the seek time formulation used in HDDs, as

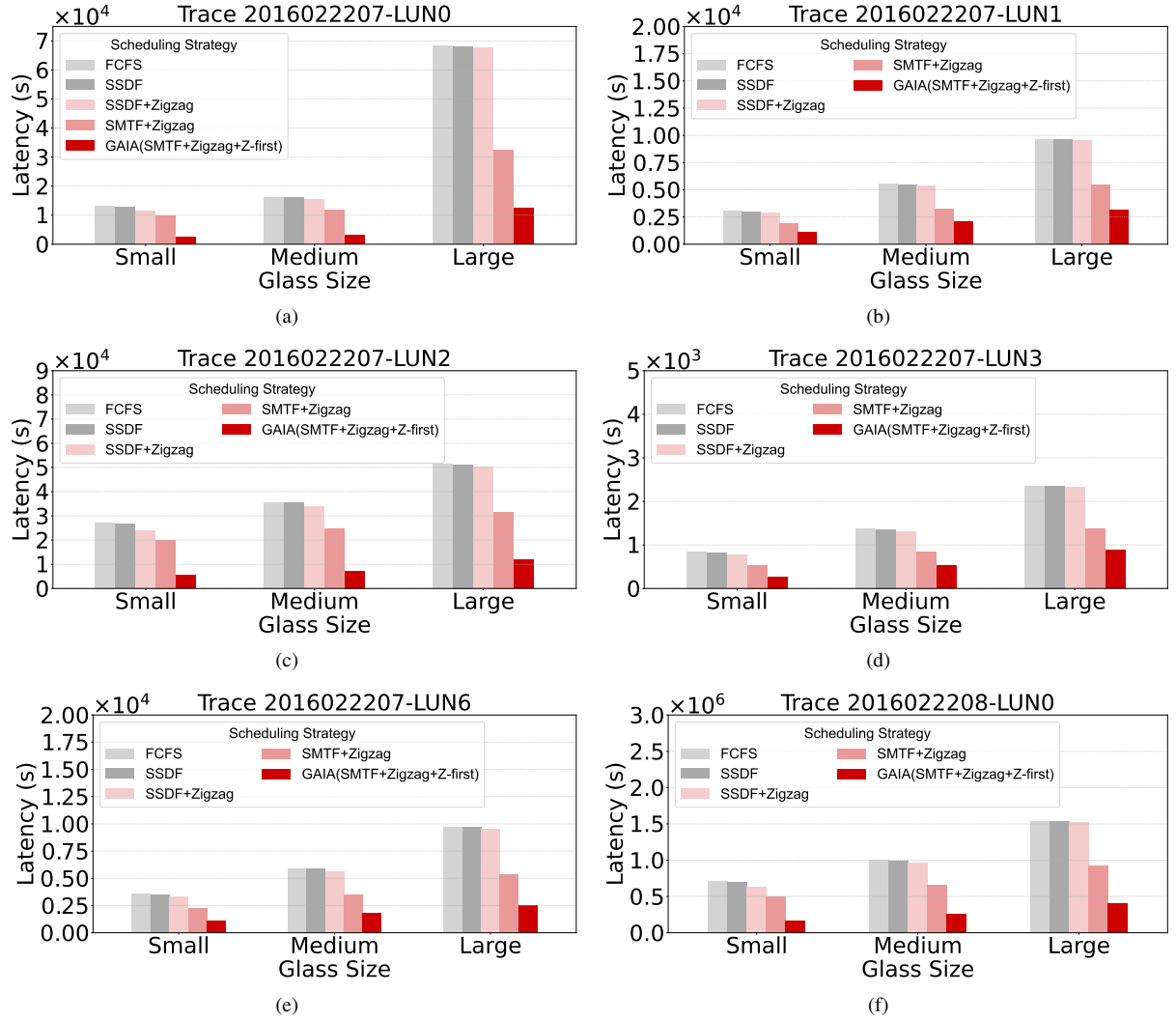


Fig. 5. Evaluation on Total Requests Latency Time for different glass sizes and traces.

discussed in [37]. When the movement distance d is smaller than a predefined threshold m , the latency is modeled based on uniformly accelerated motion, where the relationship between time and distance follows $t \propto \sqrt{d}$. For larger distances ($d \geq m$), the device is assumed to reach its maximum speed, and the latency increases linearly with distance. The complete formulation is given as:

$$f_{\text{seek}}(d) = \begin{cases} p + q\sqrt{d}, & \text{if } d < m \\ r + sd, & \text{if } d \geq m \end{cases}$$

C. Evaluation Results

Figure 5 presents the simulation results of executing various scheduling strategies on the glass storage system. The horizontal axis represents three different glass sizes, while each subplot corresponds to Fujitsu traces [35] as we mentioned in IV-A. In particular, Figures 5(a)-5(f) illustrate the results for traces 2016022207-LUN0, 2016022207-LUN1,

2016022207-LUN2, 2016022207-LUN3, 2016022207-LUN6, and 2016022208-LUN0, respectively. The vertical axis denotes the total latency incurred when processing all read requests within each trace using a given strategy. A lower latency value directly reflects better performance of the scheduling strategy. Moreover, a larger percentage reduction in latency further indicates a greater effectiveness in minimizing read delays. For all simulations in Figure 5, the request queue size was fixed to accommodate 10 concurrent read requests.

In Figure 5(a)-5(f), across all four subfigures, experimental results show that the performance difference between FCFS and SSDF is less than 1% when the glass size is medium or large. This indicates that the intuitively applicable SSDF strategy offers negligible benefits in these scenarios, highlighting the inapplicability of conventional scheduling approaches to glass storage systems. Efficient read strategies must be designed with the unique physical characteristics of glass in mind. Only in the small glass configuration does SSDF

yield a marginal improvement of 1%–3% over FCFS, due to reduced interference from long-distance movements. However, the overall improvement remains minimal and insufficient for practical performance gains.

SSDF + Zigzag strategy achieves an improvement of 6%–11% over FCFS when the glass size is small. This improvement drops to 2%–5% for medium glass, and further diminishes to only 1%–2% for the large glass. This trend is attributed to the Zigzag strategy being particularly effective in mitigating the excessive repositioning caused by long read requests that reach the tail end of the glass. Such conditions are more likely to occur in smaller glass configurations due to their higher frequency of edge traversal, which in turn makes the impact of Zigzag more pronounced. Consequently, the smaller the glass size, the greater the performance gain observed for Zigzag method.

When using FCFS as the normalization base, the SMTF + Zigzag strategy reduces total latency by 12%–39% more than the SSDF + Zigzag strategy for small and medium glass sizes. This improvement becomes even more significant for the large glass configuration, reaching a 37%–51% additional reduction in latency. These results demonstrate that SMTF + Zigzag, which prioritizes the selection of read requests based on the shortest physical movement time in glass, effectively improves read performance by accounting for acceleration-aware seek behavior. Moreover, the performance gains become increasingly pronounced as the glass size grows. Given that modern cloud storage systems are typically geared toward storing large volumes of data, larger glass media are more likely to be adopted in practice. This highlights the potential of the SMTF + Zigzag strategy as a scalable and efficient scheduling approach for next-generation glass storage systems.

If Z-axis First Strategy is incorporated at the data placement stage during the writing process, the resulting data layout can be leveraged to further improve read performance. As demonstrated in Figure 5, experimental results show that this strategy consistently yields significant latency reductions across all glass sizes, achieving improvements in the range of 62% to 82%. This confirms the effectiveness of aligning physical data placement with the access characteristics of the optical focus mechanism in glass storage systems.

In addition, all trace groups presented in Figure 5 exhibit a consistent trend, further validating that the proposed SMTF + Zigzag + Z-axis First Strategy provides substantial improvements in read performance for glass storage systems. Across the evaluated configurations, this combined strategy achieves up to 82% read latency reduction, highlighting its effectiveness and robustness across varying workload patterns and glass sizes.

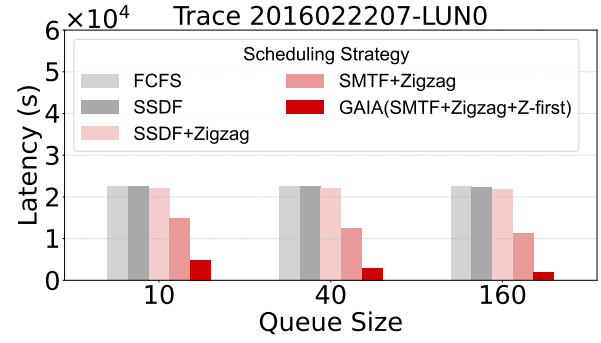


Fig. 6. Evaluation on total requests latency time for different queue sizes.

V. CONCLUSION

Quartz glass represents a transformative opportunity in long-term data preservation due to its exceptional durability and storage density. However, its mechanical-optical access model introduces novel challenges that render traditional I/O scheduling strategies ineffective. In this paper, we presented GAIA, a Glass-Aware I/O middlewAre that addresses these challenges through a physically informed scheduling and placement architecture.

GAIA comprises three core strategies: a zigzag data placement layout to streamline planar movement, a Z-axis first strategy that leverages the rapid focal tuning of the optical read mechanism, and the Shortest Moving Time First (SMTF) scheduler, which reduces latency by considering stage acceleration and direction-switching overheads. Together, these techniques create a scheduling pipeline tailored to the physical constraints and strengths of quartz glass systems.

Through extensive simulations based on real-world enterprise traces and a variety of glass sizes, we demonstrated that GAIA substantially outperforms conventional schedulers such as FCFS and SSDF, achieving up to 82% latency reduction. As demand for ultra-durable archival storage continues to rise, GAIA lays the groundwork for practical deployment of glass-based storage systems in data centers, cloud infrastructure, and scientific repositories. Looking ahead, we envision extending GAIA to support dynamic workload adaptation and hybrid glass-HDD environments, further bridging the gap between emerging storage media and system-level optimization.

VI. ACKNOWLEDGEMENT

This work was partially supported in part by the NSF Awards 2204656 and 2412437 and the National Science and Technology Council under grant nos. 114-2221-E-A49-156-MY3, 114-2628-E-A49-006, 113-2640-E-A49-012, 114-2927-I-002-525, 114-2927-I-002-532, 114-2223-E-002-011, 114-2221-E-002-219-MY3, 114-2221-E-002-222-MY3 and Ministry of Education under grant no. 114L900903 and Yushan Young Fellow Program.

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